

Shrioma Pal

Full Stack Developer | AI & Reinforcement Learning

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PROFILE

Full-stack developer specializing in AI-driven systems and reinforcement learning environments, with experience building scalable backend systems and data-driven, adaptive applications. Finalist at Meta PyTorch OpenEnv Hackathon (Top 2.8% of 31,000+ teams). Built and deployed real-world applications including an RL-based hospital triage system and AI-powered mental wellness platform, focusing on optimization, system design, and data-driven decision-making.

WORK EXPERIENCE

Labdix Pvt. Ltd., <i>Web Developer Intern</i>	03/2026 – 05/2026 Pune
- Engineered scalable backend services using Node.js, Prisma ORM, and TypeScript, supporting high API traffic - Established secure authentication and user management using JWT for controlled access - Built and optimized 15+ REST API endpoints for LMS platform - Architected backend system design and video streaming pipeline (HLS) for seamless performance - Enhanced debugging, validation, and overall system reliability, reducing errors - Contributed to production-level LMS features used by active users	

EDUCATION

B.Tech in Computer Science & Engineering, <i>Supreme Knowledge Foundation, Hooghly, WB</i>	06/2027 Hooghly, WB
CGPA: 7.95 / 10.0	
Diploma in Computer Science & Technology, <i>Acharyya Prafulla Chandra Ray Polytechnic, Jadavpur, Kolkata</i>	10/2021 – 07/2024 Jadavpur, Kolkata
OGPA: 8.9 / 10.0	

TECHNICAL SKILLS

Languages Python, Java, C++, JavaScript, TypeScript	Frontend React.js, Next.js, HTML, CSS, Tailwind CSS	Backend Flask, Node.js, Express.js, REST APIs, FastAPI, Prisma ORM	Database MongoDB, SQLite
AI/ML LangChain, FAISS, Reinforcement Learning (PPO), OpenEnv, Prompt Engineering, AI Model Integration	Tools & DevOps Git, GitHub, Docker, Postman, Vercel, Render	Core Concepts Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, System Design, Debugging, SDLC	Soft Skills Problem Solving, Team Collaboration, Communication, Adaptability, Analytical Thinking

ACHIEVEMENTS

Finalist — Meta Pytorch OpenEnv Hackathon (Top 2.8% of 31,000+ teams), <i>Meta</i>	26/04/2026
Finalist — Infosys Global Hackathon (Top 1.6% Top 33 teams among 1,942), <i>Infosys</i>	04/08/2025
Finalist — Build With Gemini Hackathon, <i>Delhi University</i>	01/10/2025

PROJECTS

MedFlow-OpenEnv, <i>RL-based Hospital Triage Environment</i>	04/2026 – Present
- Tech Stack: Python, FastAPI, Reinforcement Learning, OpenEnv, Hugging Face - Engineered an RL-based decision system improving patient prioritization, reducing critical wait times by ~30%, and outperforming baseline policies by ~25% through optimized reward-driven training - Simulated 1,000+ patient scenarios across multiple difficulty levels using a scalable OpenEnv environment with Gym-style APIs (reset, step, state) for training and evaluation - Integrated backend APIs with Hugging Face Spaces, enabling live simulation, testing, and deployment of RL-driven patient prioritization workflows	
KisanMitra, <i>Intelligent Agricultural Advisory & Marketplace</i>	07/2025 – Present
- Tech Stack: ReactJS, Flask, Python, MongoDB, Docker, Vercel, Render, JavaScript - Launched and deployed a scalable full-stack web application for agricultural advisory and marketplace - Increased user engagement by 40%, with adoption by 1,000+ users - Built and optimized 10+ REST API endpoints using Flask for real-time data processing	
MindScape, <i>AI-Powered Mental Wellness Platform</i>	05/2025 – Present
- Tech Stack: React.js, Flask, Python, DeepFace, OpenCV, LangChain, FAISS, Groq API, SQLite - Created an AI-driven recommendation system and chatbot (Keshava) serving 500+ users with a responsive mood tracking dashboard and personalized content - Developed a RAG pipeline using LangChain and FAISS, indexing 1,000+ data points for contextual recommendations - Processed real-time facial emotion detection with ~85% accuracy using DeepFace and OpenCV, correlating emotional patterns with tracked user behavior via Chrome extension	